

**DEPARTMENT OF COMPUTER & INFORMATION SYSTEMS ENGINEERING
BACHELORS IN COMPUTER SYSTEMS ENGINEERING**

Course Code and Title: CS-432 Distributed Computing

Complex Engineering Activity

BE Batch 2022, Spring Semester 2026

TERM PROJECT

Course Learning Outcome

CLO-1: Examine existing techniques for distributed computing. (Taxonomy level C3, PLO-2 Problem Analysis).

CLO-2: Analyse distributed computing solutions for their pros and cons. (Taxonomy level C4, PLO-4 Investigation).

CLO-3: Analyse distributed computing scenarios for environment friendliness and/or sustainability. (Taxonomy level C4, PLO-7 Environment and Sustainability)

Description

In this assignment, you are tasked as Computer Systems Engineer to design and implement a distributed computing system, you would also have to explore different techniques and platforms before implementation, so that you may come up with a sound and technically correct design. You may choose one of the following options for the project.

- Distributed E-Commerce System:

Design and implement a simplified distributed e-commerce platform including:

- User Service
- Product Service
- Order Service
- Payment Simulation Service
- Separate databases (or logical separation)

- Distributed File Storage System:

Design a system supporting:

- File upload/download
- Replication across nodes (at least 3 nodes)
- Separate databases (or logical separation)
- Access transparency

- Instructor-Approved Proposal

Students may propose their own distributed system provided it includes:

- Minimum 3 distributed components
- Network communication
- Clear architectural style

Students must communicate the project topic by week 09.

Deliverables

1. Design & Analysis Document

Design document must have the following sections.

- Problem Statement: Explaining real world motivation and why distributed solution is needed.
- Architectural Style: Discussing what architecture was chosen? And what alternatives were rejected?
- System Architecture Diagram
- Sustainability & Environmental Analysis

2. Working Prototype

All members must demonstrate the working prototype of the project.

Timeline

Week	Milestone
07	Project Announced
09	Proposal Submission via Google Form
12 and 13	Theory Teacher Evaluation
14	Practical Teacher Evaluation

Mandatory Requirements

- Must use network communication
- No monolithic single-process systems
- Minimum 3 distributed components
- Must include sustainability analysis
- All members must contribute

Some Extension Ideas

Student may implement the following to further enhance the project. Bonus marks will be considered on these as well.

- Docker containerization
- Cloud deployment (AWS/Azure/GCP)
- Load testing with metrics, with performance benchmarking
- Edge simulation

Other Instructions

- Students may work in groups of up to four.
- Students may add additional features to make their project distinct.

Academic Integrity and Use of AI Tools

Students may use AI tools for assistance; however:

- All design decisions must be understood and defensible.
- Students are responsible for explaining all submitted code and results.
- Inability to explain the work during the viva may result in loss of marks.

Submission and Grading

- The project shall be graded for **20 marks under the Final Practical Examination** and **10 marks under the Theory**.
- Evaluation will be conducted using the attached rubrics (available on Google Classroom) as per the announced evaluation schedule.
- All group members must be present during the submission and viva for theory. After final viva and in person submission, all group members must also submit the report and coding files on google classroom assignment as well.